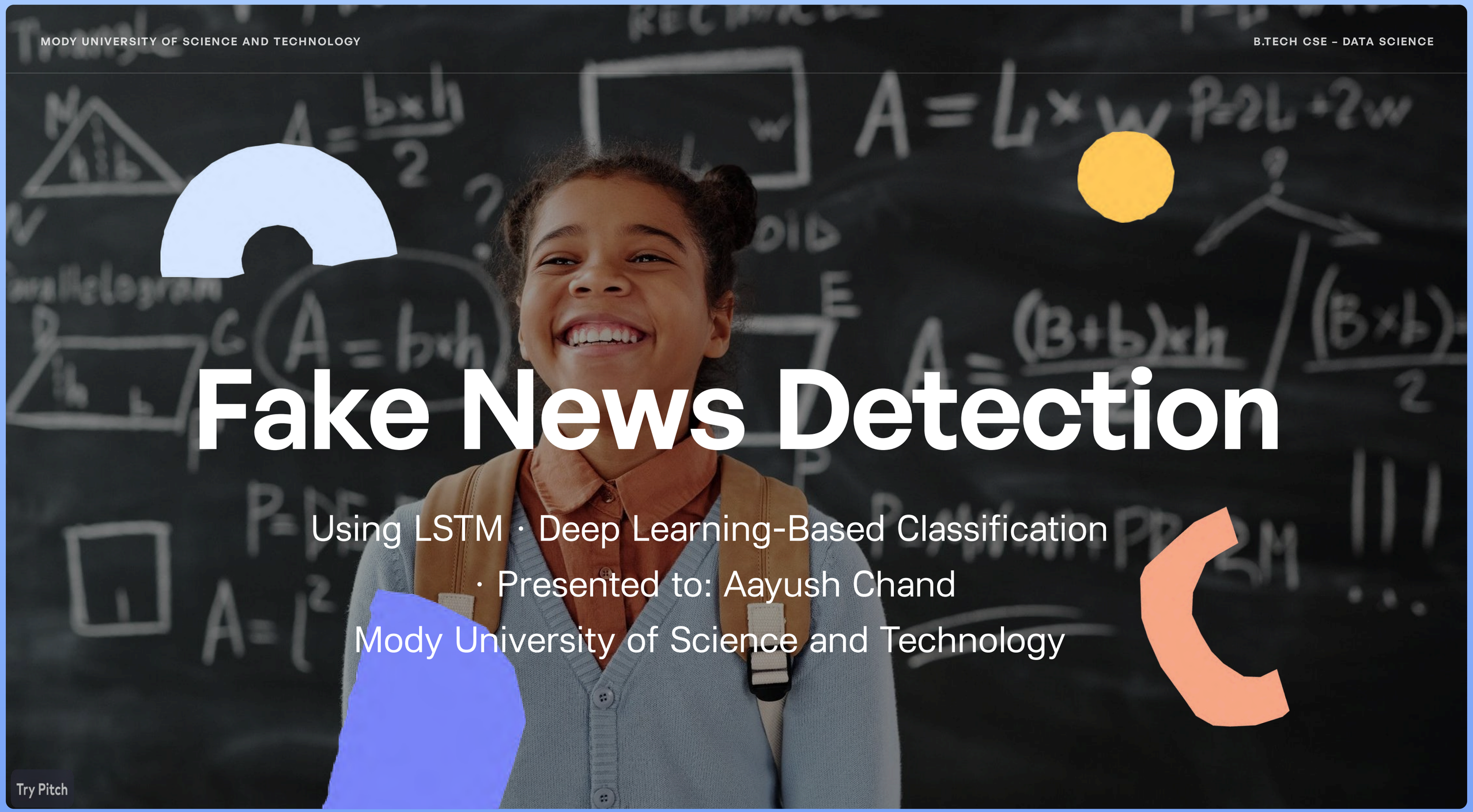


Fake News Detection

Using LSTM · Deep Learning-Based Classification

· Presented to: Aayush Chand

Mody University of Science and Technology

Try Pitch

Introduction

Online news spreads rapidly through social media and websites.

False information can mislead readers.

Manual verification of every article is time-consuming.

AI can help automate news classification.

Try Pitch

Problem Statement

The problem

Real and fabricated news are hard to tell apart, and large volumes of text need automated analysis.

Objective

Develop an LSTM-based binary text classification model that learns language patterns from labeled examples.

Try Pitch

Dataset Description

true.csv — real news articles, labeled 0.

fake.csv — fake news articles, labeled 1.

Pandas combines both datasets into one DataFrame.

Class distribution and text lengths are explored.

Try Pitch

Data Preprocessing

✔ Explore title and text lengths

✔ TextVectorization layer converts words to token IDs

✔ Vocabulary limit: 10,000 tokens

✔ Sequence length: 1,024 tokens

✔ Pad shorter sequences

✔ Fixed-length input for the network

Try Pitch

Proposed Model: LSTM

LSTM

A recurrent neural network that uses memory to learn sequential relationships and context in text.

Bidirectional LSTM + Embedding

Reads text in both directions to capture context from both sides. The embedding layer gives each word a 64-dimensional representation.

Try Pitch

Model Architecture



Sigmoid score near 0 → Real news · near 1 → Fake news (0.5 threshold)

Try Pitch

Training and Evaluation

Data split: 80:20 train-validation, stratified, random state 42

Training: 10 epochs, batch size 256

Optimizer: Adam, learning rate 0.001
Loss: Binary Cross-Entropy

Metrics: Accuracy and AUC

Try Pitch

Results and Discussion

~99.8–99.9% validation accuracy

Very high on the validation split, but this doesn't guarantee the same accuracy on new, real-world news.

Try Pitch

Conclusion & Future Scope

Conclusion

Built a fake-news classifier using NLP, word embeddings and Bidirectional LSTM, evaluated with accuracy and AUC.

Future scope

- Test on independent, unseen datasets
- Add source credibility and fact-checking
- Build a web application
- Monitor changing misinformation patterns

Try Pitch

Thank you

Questions? — Piya Sachdeva

Try Pitch

Pitch

Want to make a presentation like this one?

Start with a fully customizable template, create a beautiful deck in minutes, then easily share it with anyone.

Create a presentation (It's free)